

GLUCOSE DETERMINATION

(GOD/POD METHOD)

PRINCIPLE

Glucose is oxidized to gluconic acid and hydrogen peroxide in the presence of glucose oxidase (GOD.) Hydrogen peroxide further reacts with phenol and 4-aminoantipyrine by the catalytic action of peroxidase (POD) to form a red-colored quinonimine dye complex. Intensity of the color formed is directly proportional to the amount of glucose present in the sample.



REAGENT COMPOSITION

L1: Enzyme reagent containing phosphate buffer (pH 7.5), 4-aminoantipyrine, phenol, GOD, POD, *mutarotase and stabilizers

S: Glucose standard (100mg/dL)

SAMPLES

Serum, plasma or CSF. Glucose is reported to be stable for 7 days when stored at 2-8°C.

INTERFERENCES

The POD reaction is much less specific than the GOD reaction. Various substances such as uric acid, ascorbic acid, bilirubin, hemoglobin, glutathione and tetracycline inhibit the reaction (presumably by competing with the chromogen 4-aminoantipyrine for H_2O_2), producing lower values. The urine contains high concentrations of substances that interfere with the peroxidase reaction (such as uric acid), producing falsely low results. The glucose oxidase method therefore should not be used for urine.

PROCEDURE

Wavelength/filter 505 nm

Temperature 37°C/Room temperature

Light path 1 cm

Pipette into clean dry test tubes labeled as blank (B), standard (S) and test (T):

<i>Addition sequence</i>	<i>B (μl)</i>	<i>S(μl)</i>	<i>T(μl)</i>
<i>Glucose reagent (L1)</i>	1000	1000	1000
<i>Distilled Water</i>	10	-----	-----
<i>Glucose standard (S)</i>	-----	10	-----
<i>Sample</i>	-----	-----	10

Mix well and incubate at 37°C for 5 min. or at room temperature (25°C) for 10 min. Measure the absorbance of the standard (Abs. S), and test sample (Abs. T) against the blank, within 60 minutes.

CALCULATIONS

$$C=100 \times (\text{Abs. T}/\text{Abs. S}) [\text{mg/dl}] \text{ or}$$

$$C=5.55 \times (\text{Abs. T}/\text{Abs. S}) [\text{mmol/l}]$$

LINEARITY

The test is linear up to a glucose concentration of 500 mg/dl. If values exceed this limit, dilute the serum with normal saline (NaCl 0.9%) and repeat the assay, Calculate the value using the proper dilution factor.

NORMAL REFERENCE VALUES

Serum/plasma: Fasting	70-100mg/dl
(2 hrs. post-prandial)	upto 150 mg/dl
CSF:	50-80 mg/dl

It is recommended that each laboratory establish its own normal range representing the target population.

NOTE

To avoid glycolysis, the serum should be separated from the clot as soon as possible and the plasma

should be collected in an EDTA+sodium fluoride vial (0.5 mg+1mg per/ml of blood)

CLINICAL SIGNIFICANCE

Generally, *fasting plasma glucose (FPG)* ≥ 126 mg/dl (7mmol/l) on more than one occasion are diagnostic of diabetes mellitus. The diagnosis of most cases of diabetes mellitus is established with this criterion. However, some investigators believe that hyperglycemia may be a relatively late development in the course of the type 2 diabetes, delaying the diagnosis and underestimating the prevalence of diabetes mellitus in the population. The onset of type 2 diabetes probably occurs at least 4 to 7 years before clinical diagnosis.

The *oral glucose tolerance test (OGTT)* is more sensitive than fasting glucose test early in the course of type 2 diabetes. The OGTT test is carried out in the morning. First, plasma glucose is measured fasting, then every 30 minutes for 2 hours after an oral glucose load. For non-pregnant adults, the recommended load is 75g. Blood glucose concentrations ≥ 200 mg/dl at the end of the two hours are indicative of diabetes mellitus.

The measurement of *glycated-hemoglobin (Ghb)* has long been used in the management of established diabetes as a biomarker of long-term glycemic control. This test has recently been recommended for use in the diagnosis of diabetes, with a threshold level of 6.5%. Diabetic patients, especially those who need insulin therapy, require careful monitoring to maintain control of blood glucose. Patients may use portable meters for measurement of blood glucose concentrations and modify their insulin dose based on this glucose value. Simple test strips are available that permit rapid and reasonably accurate measurements on a drop of whole blood.

Major Diagnostic Criteria for Diabetes and Prediabetic or At-Risk States.				
Measure	American Diabetes Association		World Health Organization	
	Diabetes	Prediabetes	Diabetes	Impaired Glucose Regulation
Fasting plasma glucose	≥ 126 mg/dl	100–125 mg/dl (IFG)	≥ 126 mg/dl	110–125 mg/dl (IFG)
2-Hr plasma glucose (during an OGTT with a loading dose of 75 g)	≥ 200 mg/dl	140–199 mg/dl (IGT)	≥ 200 mg/dl	140–199 mg/dl (IGT)
Casual (or random) plasma glucose (in a patient with classic hyperglycemic symptoms)	≥ 200 mg/dl		≥ 200 mg/dl	
Glycated hemoglobin	$\geq 6.5\%$	5.7–6.4%	$\geq 6.5\%$	

TOTAL CHOLESTEROL

PRINCIPLE

This method for the measurement of total cholesterol in serum involves the use of three enzymes: cholesterol esterase (CE), cholesterol oxidase (CO) and peroxidase (POD). In the presence of peroxidase the mixture of phenol and 4-aminoantipyrine (4-AA) are condensed by hydrogen peroxide to form a quinoneimine dye proportional to the concentration of cholesterol in the sample.

Cholesteryl Esters CE Cholesterol+Fatty Acids

Cholesterol + O₂ CO Cholestenone + H₂O₂

4-AA+ Phenol H₂O₂, POD Quinoneimine + 4H₂O

REAGENT COMPOSITION

R1: Buffer PH 7.0, CE, CO, POD, 4-AA, phenol, biocides.

CAL: Cholesterol standard (200 mg/dl or 5.18 mmol/l)

SAMPLES

Serum, EDTA or heparinized plasma free of hemolysis. Cholesterol in serum or plasma is stable up to 5 days at 2-8 °C and for a few months at -20 °C.

INTERFERENCES

This method is subject to interference from substances that compete in the oxidase and peroxidase reactions like bilirubin, hemoglobin, ascorbic acid and other drugs and substances.

PROCEDURE

Wavelength/filter 510 nm

Temperature 37°C/ Room temperature

Light path 1 cm

Bring reagents and samples to room temperature. Pipette into clean dry test tubes labeled as blank (B), standard (S) and test (T):

Addition sequence	B (μl)	S(μl)	T (μl)
R1	1000	1000	1000
Distilled water	10	-----	-----
CAL	-----	10	-----
Sample	-----	-----	10

Mix and incubate the tubes 10 minutes at room temperature or 5 minutes at 37°C. Measure the

absorbance of the standard (Abs. S), and test sample (Abs. T) against the blank. The color is stable for at least 30 minutes protected from light.

CALCULATIONS

$$C=200 \times (\text{Abs. T}/\text{Abs. S}) [\text{mg/dl}] \text{ or}$$

$$C=5.18 \times (\text{Abs. T}/\text{Abs. S}) [\text{mmol/l}]$$

LINEARITY

Samples with concentrations higher than 600 mg/dL should be diluted 1:2 with saline and assayed again. The results should then be multiplied by 2.

REFERENCE VALUES

Updated clinical values of total cholesterol used to classify risk groups.

Total Cholesterol	Risk Classification
<200 mg/dL (5.18 mmol/L)	Desirable
200-239 mg/dL (5.18 - 6.2mmol/L)	Borderline high
> 240 mg/dL(> 6.2 mmol/L)	High

CLINICAL SIGNIFICANCE

Cholesterol exists in the human blood as a free sterol and in an esterified form. The knowledge of the plasma level of lipids (cholesterol and triglycerides) together with lipoproteins of high and low density (HDL and LDL) aids in the detection of many conditions bound to metabolic disorders of high risk. The imbalance in the level of lipoproteins in plasma leads to hyperlipoproteinemias, a group of disorders that affects lipid level in serum, causing coronary heart disease (CHD) and atherosclerosis, conditions in which the cholesterol levels are important tools in their diagnosis and classification.

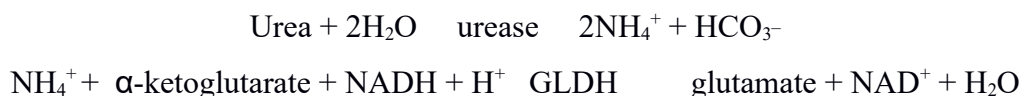
Jaundice of the obstructive type usually is accompanied by elevated total serum cholesterol with a normal ester fraction. Diabetes, hypothyroidism, and certain types of kidney disease are other disorders that may exhibit the same cholesterol disturbance.

Low total cholesterol values with normal ester fractions are noted mainly in hyperthyroidism and malnutrition.

UREA

PRINCIPLE

Urea is hydrolyzed in the presence of water and urease to produce ammonia and carbon dioxide. In the presence of glutamate dehydrogenase (GLDH) the formed ammonium ion reacts with α -ketoglutarate and NADH to form glutamate and NAD^+ . An equimolar quantity of NADH undergoes oxidation during the reaction and therefore the decrease in absorbance at 340 nm due to NADH oxidation is directly proportional to the urea concentration in the sample.



REAGENT COMPOSITION

L1: Enzyme reagent

L2: Substrate reagent

S: Urea standard (40mg/dl)

WORKING REAGENT PREPARATION

Mix 4 parts of L1 with 1 part of L2.

SAMPLES

Serum, plasma and urine. Urine should be diluted 1+49 with distilled water before the assay; the results must then be multiplied by 50. Urea is reported to be stable in the serum for 5 days when stored at 2-8°C.

INTERFERENCES

Interference may be expected from endogenous ammonia.

PROCEDURE

Wavelength 340 nm (kinetic method)

Temperature 37°C

Light path 1 cm

Pipette into clean dry test tubes labeled as standard (S) and test (T):

Addition sequence	S(μl)	T (μl)
Working reagent	1000	1000
Urea standard	10	-----
Sample	-----	10

Mix well and start the stopwatch. After 30 seconds at 37°C, record the first absorbance (A1) of the standard and the test. Exactly after 60 seconds, record the second absorbance (A2).

CALCULATIONS

$$\text{Urea in mg/dl} = \frac{A1-A2 \text{ of T}}{A1-A2 \text{ of S}} \times 40$$

$$\text{Urea nitrogen in mg/dl} = \text{Urea in mg/dl} \times 0.467$$

LINEARITY

The procedure is linear upto 250 mg/dl. If values exceed this limit, dilute the serum with normal saline (NaCl 0.9%) and repeat the assay. Calculate the value using the proper dilution factor.

NORMAL REFERENCE VALUES

Serum/plasma urea	14-40mg/dl
Urine urea:	20g/l in 24 hrs

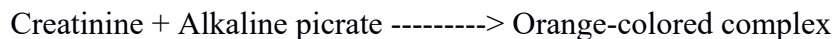
CLINICAL SIGNIFICANCE

Measurement of blood and urine urea is a useful indicator of kidney function. However, a number of extrarenal factors may influence the circulating urea concentration limiting its value as a test of kidney function. For example, plasma urea concentration can be increased by high-protein diet, increased protein catabolism, dehydration, treatment with cortisol or decreased perfusion of the kidneys due to heart failure. Low urea levels may be indicative of acute tubular necrosis, low protein intake or severe liver disease. Urea measurement, when done in conjunction with creatinine test, gives a clearer picture of the status of the kidneys.

CREATININE

PRINCIPLE

Creatinine under alkaline conditions reacts with picrate ions forming an orange-colored complex. The formation rate of the complex measured through the increase of absorbance in a fixed interval of time is proportional to the concentration of creatinine in the sample.



REAGENT COMPOSITION

L1: Picric Acid

L2: Alkaline buffer

S: Creatinine standard (2mg/dl)

SAMPLES

Serum or urine. Creatinine is stable in serum for 1 day at 2-8°C. Creatinine from random samples of urine is stable for 4 days at 2-8 °C. Urine of 24 hours collection is preferred. Dilute the specimen 1:50 with distilled water before the assay.

PROCEDURE

Wavelength 520 nm

Temperature 37°C

Light path 1 cm

Pipette into clean dry test tubes labeled as standard (S) and test (T):

Addition sequence	S(μl)	T (μl)
L1	500	500
L2	500	500
Bring reagents to the assay temperature and add		
Standard	100	-----
Sample	----	100

Mix well and read the initial absorbance A_1 for the standard and test exactly after 30 seconds. Read another absorbance A_2 of the standard and test exactly 60 seconds later. Calculate the change in absorbance ΔA for both.

$$\text{For standard } \Delta AS = A_2S - A_1S$$

For test $\Delta AT = A_2T - A_1T$

CALCULATIONS

Serum creatinine in mg/dl = $\Delta AT / \Delta AS \times 2$

Urine creatinine in g/l = $\Delta AT / \Delta AS \times 1$

Urine creatinine in g/24hrs = Urine creatinine in g/l X vol of urine in 24 hrs.

LINEARITY

The procedure is linear upto 20mg/dl of creatinine. If values exceed this limit, dilute the sample with distilled water and repeat the assay. Calculate the value using the proper dilution factor.

NORMAL REFERENCE VALUES

	Serum	Urine in 24 hrs. collection
Males	0.6-1.2 mg/dl	1.1-3 gm
Females	0.5-1.1 mg/dl	1.0-1.8 gm

NOTE

The determination is not specific and may be affected by the presence of large quantities of reducing substances. As the test is temperature-sensitive, it is essential to maintain the indicated reaction timings and temperatures carefully during the test procedure.

CLINICAL SIGNIFICANCE

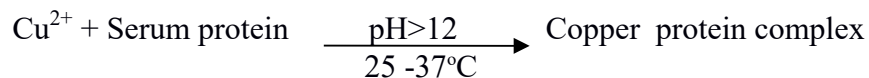
The daily production of creatinine depends on muscle mass and is excreted out of the body entirely by the kidneys. The level of creatinine in healthy individuals is independent of diet and thus relatively constant. The creatinine clearance (CC) test is one of the most sensitive tests to diagnose renal function especially the glomerular filtration rate (GFR) because the concentration of creatinine in serum is dependent almost entirely upon its rate of excretion by the kidney.

Elevated levels of creatinine are usually seen in renal dysfunction and reduced renal blood flow (shock, dehydration, congestive heart failure.) Decreased levels are seen in muscular dystrophy.

TOTAL PROTEIN

PRINCIPLE

In the *biuret reaction*, a chelate is formed between the Cu^{2+} ion and the peptide bonds of the proteins in alkaline solutions to form a violet colored complex whose absorbance is measured photometrically. The intensity of the color produced is proportional to the concentration of protein in the sample.



REAGENT COMPOSITION

R1: Biuret reagent

CAL: Protein standard (8 g/dL).

SAMPLES

Serum or heparinized plasma. Total protein is stable in serum and plasma for 1 week at room temperature, for at least 1 month refrigerated at $2-8^{\circ}\text{C}$, and for up to 2 months at 20°C

INTERFERENCES

Bilirubin and hemoglobin above certain concentrations cause significant interference. Lipemia causes negligent interference.

PROCEDURE

Wavelength 540 nm

Temperature 37°C

Light path 1 cm

Pipette into labeled tubes:

	Blank	Sample	CAL (Standard)
R1 (Biuret)	1.0 mL	1.0 mL	1.0 mL
Sample	—	20 μL	—
CAL (Standard)	—	—	20 μL

Mix and incubate the tubes for 10 minutes at 37°C . Read the absorbance (A) of the samples and

the standard at 540 nm against the reagent blank. The color is stable for at least 1 hour.

CALCULATIONS

$$\text{Concentration of Test} = A_T / A_S \times 7 \text{ g/dL}$$

REFERENCE VALUES

Serum

Adults	6.6 – 8.7 g/dL
Prematures	3.6 – 6.0 g/dL
Newborns	5.3 – 8.9 g/dL
Pregnancy	Concentration lowers from 6.9 to 6.1 g/L

Total serum protein is higher by 0.4 to 0.8 g/dL with the subject supine than with the subject ambulatory.

Plasma

Plasma protein is 0.2 to 0.4 g/dL higher due to the presence of fibrinogen in the sample.

CLINICAL SIGNIFICANCE

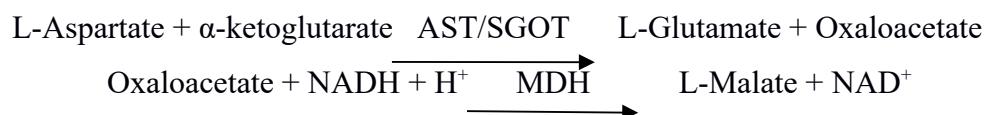
The total protein is measured in serum to give an indication of total immunoglobulin concentration since $([\text{total protein}] - [\text{albumin}]) = [\text{globulins}]$ of which the major component is immunoglobulins. The total protein is sometimes included in the liver function tests; some chronic liver diseases cause increases in [immunoglobulins], which increases total protein (though this may be offset by a decrease in [albumin]) Total protein should be measured in suspected humoral immunodeficiency, for which the definitive diagnosis is by measurement of individual immunoglobulins.

Total protein measurement is of little value without simultaneous measurement of albumin. And in many cases the specific types of immunoglobulins should also be quantified.

ASPARTATE AMINOTRANSFERASE (AST/SGOT)

PRINCIPLE

Aspartate aminotransferase (AST/SGOT) catalyzes the transfer of the amino group from aspartate to α -ketoglutarate with the formation of glutamate and oxaloacetate. The latter is reduced to malate by malate dehydrogenase (MDH) in the presence of reduced nicotinamide adenine dinucleotide (NADH). The reaction is monitored kinetically at 340 nm by the rate of decrease in absorbance resulting from the oxidation of NADH to NAD^+ proportional to the activity of AST present in the sample.



REAGENT COMPOSITION

R1: AST substrate

R2: AST coenzyme

REAGENT PREPARATION

Working reagent: Mix 4mL of R1 + 1mL of R2. Should be protected from light.

SAMPLES

Serum, EDTA or heparinized plasma free of hemolysis. AST is stable in serum or plasma 24 hours at room temperature and for 1 week at 2 - 8 °C

PROCEDURE

Wavelength 340 nm

Temperature 30/37 °C

Light path 1 cm

Preincubate working reagent, samples and controls to reaction temperature. Pipette into a cuvette:

Reaction temperature	37°C	30°C
Working reagent	1.0 mL	1.0 mL
Sample or control	50 μL	100 μL

Mix gently by inversion. Insert cuvette into the cell holder and start stopwatch. Incubate for 1

minute and record initial absorbance reading. Repeat the absorbance readings exactly after 1, 2 and 3 minutes. Calculate the difference between absorbances. . Calculate the mean of the results to obtain the average change in absorbance per minute ($\Delta A/\text{min}$).

CALCULATIONS

$$U/L = \Delta A/\text{min} \times 3333 \text{ (37}^\circ\text{C)}$$

$$U/L = \Delta A/\text{min} \times 1746 \text{ (30}^\circ\text{C)}$$

REFERENCE VALUES

Serum, plasma

Adults	37°C	Up to 40 U/L (0.67 $\mu\text{kat/L}$)
	30°C	Up to 25 U/L (0.42 $\mu\text{kat/L}$)

CLINICAL SIGNIFICANCE

The groups of enzymes called transaminases exist in tissues of many organs. Necrotic activity in these organs causes a release of abnormal quantities of enzyme into the blood where they are measured.

Since heart tissue is rich in AST increased serum levels appear in patients after myocardial infarction, as well as in patients with muscle disease, muscular dystrophy and dermatomyositis.

The liver is specially rich in ALT, hence this enzymes is usually measured as a test for infectious and toxic hepatitis, although high levels of both ALT and AST may also be found in cases of liver cell damage and acute pancreatitis, suggesting that the obstruction of the biliary tree by the edematous pancreas and the presence of associate hepatic disease may contribute to elevated AST levels in these patients.

ALANINE AMINOTRANSFERASE (ALT/SGPT)

PRINCIPLE

Alanine aminotransferase (ALT/SGPT) catalyzes the transfer of the amino group from alanine to α -ketoglutarate with the formation of glutamate and pyruvate. The latter is reduced to lactate by lactate dehydrogenase (LDH) in the presence of reduced nicotinamide adenine dinucleotide (NADH).

The reaction is monitored kinetically at 340 nm by the rate of decrease in absorbance resulting from the oxidation of NADH to NAD^+ through the activity of ALT present in the sample.



REAGENT COMPOSITION

R1 (ALT substrate). TRIS buffer 150 mmol/L pH 7.3, L-alanine 750 mmol/L lactate dehydrogenase > 1350 U/L

R2 (ALT coenzyme). NADH 1.3 mmol/L, α -ketoglutarate 75 mmol/L. Biocides.

REAGENT PREPARATION

Working reagent: Mix 4mL of R1 + 1mL of R2. Protect from light.

SAMPLES

Serum, EDTA or heparinized plasma free of hemolysis.

ALT is stable in serum or plasma 24 hours at room temperature and for 1 week at 2 - 8 °C.

INTERFERENCES

- Lipemia (intralipid > 15 g/L) does not interfere.
- Bilirubin (>30 mg/dL) does not interfere.
- Hemoglobin (>10 g/dL) does not interfere.
- Slight or moderate elevations of AST and ALT activities may be observed after intake of alcohol and after administration of various drugs, such as salicylates, opiates and ampicillin.

MATERIALS REQUIRED

- Photometer or spectrophotometer with thermostated cell compartment set at 30/37 °C , capable of reading at 340 nm.
- Stopwatch
- Cuvettes with 1cm path length
- Pipettes to measure reagent and samples.

PROCEDURE

1. Preincubate working reagent, samples and controls to reaction temperature (37°C or 30°C)
2. Set the photometer to zero absorbance with distilled water.
3. Pipette into a cuvette:

Reaction temperature	37 °C	30 °C
Working reagent	1.0 mL	1.0 mL
Sample	50 µL	100µL

4. Mix gently by inversion. Insert cuvette into the cell holder and start stopwatch.
5. Incubate for 1 minute and record initial absorbance reading.
6. Repeat the absorbance readings exactly after 1, 2 and 3 minutes.
7. Calculate the difference between absorbances.
8. Calculate the mean of the results to obtain the average change in absorbance per minute ($\Delta A/\text{min}$).

CALCULATIONS

$$U/L = \Delta A/\text{min} \times 3333 \text{ (37 °C)}$$

$$U/L = \Delta A \text{ min} \times 1746 \text{ (30 °C)}$$

Samples with $\Delta A/\text{min}$ exceeding 0.160 at 340 nm should be diluted 1:10 with saline and assayed again. Multiply the results by 10. If results are to be expressed as SI units apply: $U/L \times 0.01667 = \mu\text{kat/L}$

REFERENCE VALUES

Serum, plasma

Adults	37°C	Up to 40 U/L (0.67 μ kat/L)
	30°C	Up to 25 U/L (0.42 μ kat/L)

Levels approximately twice the adult level are seen in neonates and infants. These values decline to adult level by approximately 6 months of age.

Clinical Significance

1) Physiologic changes in the catalytic concentration of ALT in serum

The catalytic activity of ALT is decreased in normal umbilical blood; the concentration gradually rises to reach maximal values on day 5 postnatally.

2) Pathologic changes in the catalytic concentration of ALT in serum

A) Increased values of the catalytic concentration of ALT in serum are found in:

- active cirrhosis,
- acute extrahepatic biliary obstruction (ALT usually greatly exceeds AST),
- acute hepatitis;
- considerable increase, which is relatively greater than AST increase; ALT level is also increased in jaundice-free episodes.
- dermatomyositis,
- hepatotoxic lesions,
- infectious mononucleosis (highest ALT level in the second week, returning to normal ALT value in the fifth week of disease),
- infectious or toxic hepatitis,
- initial therapy with clofibrate,
- myocardial infarction (smaller increase in comparison to AST),
- obstructive jaundice,
- primary or metastatic liver carcinoma,
- progressive muscular dystrophy,
- relapse of liver cirrhosis,
- therapy with salicylates (plasma concentration >250 mg/L),

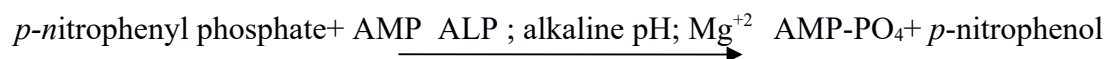
viral hepatitis.

B) Decreased values of ALT in serum are found in: vitamin B₆ deficiency

ALKALINE PHOSPHATASE (ALP)

PRINCIPLE

Alkaline phosphatase hydrolyses *p*-nitrophenyl phosphate and the phosphate is transferred to AMP (2-Amino-2-methyl-1-propanol.) The increase in absorbance at 405 nm at 37°C is measured and this is proportional to the amount of alkaline phosphatase that is present in the sample.



REAGENT CONTENT AND PREPARATION

BUF: Buffer

SUB: Substrate

Pipette 2 ml from SUB into one bottle of BUF and mix thoroughly. This working reagent must be kept protected from light

SAMPLES

Serum, heparinated plasma. Loss of activity after seven days.

INTERFERENCE

No interference of triglycerides up to 2,000 mg/dl, hemoglobin up to 250 mg/dl, and bilirubin up to 40 mg/dl.

PROCEDURE

Wavelength 405 nm

Temperature 30°C or 37°C

Light path 1 cm

Warm reagents and cuvettes up to the desired temperature. Temperature must be kept constant ($\pm 0.5^\circ\text{C}$) for the duration of the test.

Pipette into cuvettes	30°C, 37°C
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0Sample	20µl
Working reagent	1000 µl
Mix, read the absorbance after 1 minute and at the same time start the stop watch. Read the absorbance again exactly after 1, 2 and 3 minutes.	

From the readings, calculate the mean absorbance change per minute ($\Delta A/\text{min}$). Calculate the alkaline phosphatase activity in the sample using the following factor:

$$U/l = 2757 \times \Delta A/\text{min}$$

Conversion factor from traditional units (U/l) to SI-units (kat/l):

$$1 \text{ U/l} = 16.67 \times 10^{-3} \mu\text{kat/l}$$

$$1 \mu\text{kat/l} = 60 \text{ U/l}$$

REFERENCE VALUES

Serum values [U/l]		
Temperature	30°C	37°C
Women	28-78	42-98
Men	38-94	53-128

CAUTION

During the reaction p-nitrophenol is produced. This is poisonous when inhaled, swallowed or when absorbed through the skin. If the reagents come into contact with skin or mucous membranes, wash copiously with water! Contact a doctor if you feel unwell.

CLINICAL SIGNIFICANCE

Most human tissues contain ALP; kidney, liver, bone, intestine, reticuloendothelial tissue, and placenta are particularly rich sources. Enzymes derived from different tissues share catalytic properties, but differ in their chemical and physical characteristics. Various methods have been used to separate organ-specific isoforms. None are entirely satisfactory, particularly in quantifying components of the liver/bone isoenzyme group.

The majority of sustained elevated ALP levels are associated with disorders of the liver or bone, or both. Therefore, these organ systems are of prime consideration in the differential diagnosis. A variety of primary and secondary hepatic conditions may be associated with elevated serum ALP levels. Since production is increased in response to cholestasis, serum ALP activity provides a sensitive indicator of obstructive and space-occupying lesions of the liver.